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Course Summary Document

Digital Law and Data Protection

Chapter 1: Introduction & Fundamentals of Digital Law

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1. History

1.1 Libertarian Vision of Digital Law

In the early days of ARPANET and the 1990s public Web deployment, engineers and researchers viewed the Internet as a decentralized, borderless space free from state sovereignty. This philosophy was formalized in John Perry Barlow's 1996 'Declaration of the Independence of Cyberspace', which rejected government interference in the digital realm.

Usage Breakdown and Physicalization of the Network

The libertarian utopia assumed information was purely immaterial. However, the transition from Web 1.0 (static) to Web 2.0/3.0 (data-driven, commercial) highlighted three material and economic realities:

- **Physical Infrastructure:** Underlying physical infrastructure (submarine cables, satellite networks, core routers, and data centers anchored in physical national territories and subject to local laws).
- **Platform Monopolization:** The shift from a decentralized network to a market dominated by private tech monopolies:
  - GAFAM (Google, Apple, Facebook/Meta, Amazon, Microsoft) - US Tech Giants.
  - BATX (Baidu, Alibaba, Tencent, Xiaomi) - Chinese Tech Giants.

As power centralized within GAFAM/BATX, private platforms dictated their own rules over users and states, operating through:

- **Consent & Targeted Advertising:** Surveillance-based business model requiring personal data in exchange for 'free' services under take-it-or-leave-it terms.
- **Information Access:** Proprietary algorithms controlling feed visibility and search results, influencing public opinion and elections.
- **Example (Facebook & Cambridge Analytica, 2016):** In 2016, personal data of millions of Facebook users was collected without consent via a quiz app. AI algorithms processed this data to build psychographic profiles for targeted political campaigns.
- **Proliferation of Cybercrime:** The absence of early regulation led to financial fraud, industrial espionage, illicit content dissemination, and Distributed Denial of Service (DDoS) attacks.
- **Example (Mirai Botnet Attack, 2016):** In the 2010s, Internet of Things (IoT) devices expanded without legal security standards. In 2016, students created the Mirai malware, which scanned for IoT devices using default factory credentials (e.g., admin/admin), compromised over 600,000 devices into a botnet, and launched massive global DDoS attacks.

Digital Law was born at the intersection of borderless digital infrastructure and the traditional territorial principle of law (late 1990s - early 2000s). It represents the adaptation of contract, criminal, civil liability, and intellectual property laws to automated data processing.

Key Legal Precedents & Case Studies

Case Study	Jurisdiction & Date	Legal Issue	Key Legal Outcome / Adaptation
UEJF & LICRA v. Yahoo! Inc.	Paris, France v. California, USA (May 2000)	Nazi memorabilia auctioned on Yahoo! US portal accessible in France. Conflict between US 1st Amendment (free speech) and French Penal Code (prohibiting Nazi symbols).	French courts claimed jurisdiction based on local harm. Ordered Yahoo! to implement IP geoblocking and browser language filtering to restrict French users.
Gutnick v. Dow Jones & Co. Inc.	Melbourne, Australia v. New Jersey, USA (Dec 2002)	Defamatory article published on US servers (barrons.com) regarding an Australian businessman. US law requires proving 'actual malice'; Australian law is stricter.	Australian Supreme Court ruled that publication occurs where information is downloaded and read (screen display), establishing local defamation jurisdiction.
AADL Platform Hacking Case	Algiers, Algeria (April 2016)	Hackers breached the national housing platform, altering site choices for ~2,000 applicants. Digital intrusion vs. physical theft.	Algerian courts applied Penal Code Art. 323 bis (Automated Data Processing Systems). Admitted server logs and digital forensics as valid legal evidence.

1.2 Digital Sovereignty and Internet Governance

The Multi-stakeholder Governance Model

Unlike traditional telecommunications managed by state monopolies, Internet governance relies on a shared multi-stakeholder model:

- **ICANN (Internet Corporation for Assigned Names & Numbers):** Coordinates global unique identifiers, IP address allocation (IPv4/v6), and DNS root management (.dz, .com, .org).
- **IETF (Internet Engineering Task Force):** Open technical body developing Internet standards as RFCs (e.g., RFC 791 IPv4, RFC 9110 HTTP, RFC 8446 TLS 1.3).
- **ITU (International Telecommunication Union):** UN agency managing global radio spectrum allocation and telecom network cross-border interconnections.

Digital Sovereignty

Digital sovereignty refers to a state's capacity to maintain control over its digital destiny: protecting critical infrastructure, safeguarding citizen data, ensuring national security, and enforcing jurisdiction over local operators. It rests on two main pillars:

- **Technical Pillar:** Deployment of local Internet Exchange Points (IXPs) to keep national traffic internal; hosting sensitive data (health, civil registry, defense) in sovereign local data centers; ensuring network equipment security against backdoors.
- **Legal Pillar:** Mandating data localization requirements and strictly regulating cross-border data transfers to countries lacking adequate legal protection.

Sovereignty Vulnerability Examples

- **1. Algeria (Submarine Cable Outages & Overseas Hosting):** Historically, many Algerian domain names and government sites were hosted abroad. Submarine fiber optic cable cuts (e.g., SMW4 off Annaba) severely disrupted national internet access. Because local data had to route through European data centers, domestic services halted, proving that external data hosting undermines operational autonomy.
- **2. Europe (French Health Data Hub & US CLOUD Act):** France created the Health Data Hub in 2019 to centralize 67M+ health records on Microsoft Azure. Despite European servers, Microsoft is subject to the 2018 US CLOUD Act, allowing US authorities to request data regardless of location. EU courts acknowledged that GDPR protection is compromised under US extraterritorial jurisdiction.

National Models of Digital Sovereignty

- **China:** Complete self-reliance model with total state control, strict filtering, and domestic tech ecosystem deployment.
- **United States:** Global technological and economic dominance model driven by leading tech multinationals and infrastructure control.
- **Russia:** 'Sovereign Internet' (Runet) model emphasizing national security, domestic platforms (Yandex, VKontakte), autonomous national DNS, and network decoupling capabilities.

Recent Milestone Examples in Algeria (2025–2026)

- **AI Supercomputing Center:** Launched in March 2025 in Oran; equipped with GPU/TPU clusters dedicated to AI research, startups, and governmental workloads.
- **National Digital Services Center:** Inaugurated in July 2026; features twin active-active Tier III certified data centers (Mohammadia & Blida) designed to host all public administration data within national borders.